

Phase 5

Deterministic Evidence Integration and Reproducibility Analysis - Available Frozen Summary

v1.0

Phase 5 - Casual Reader Summary

What the genome research found, without the technical detail. FROZEN SCIENTIFIC RECORD - VERSION 1.0. Prepared by Zachariah P. Laing, 29 September 2026.

This is a plain-language companion to the frozen Phase 5 scientific record. It simplifies terminology but does not change the scientific conclusion.

The short version

The deletion does not look biologically meaningless. It removes or disrupts a group of genes that have real roles in the retina, brain development, nerve-cell communication, membrane biology, cellular cleanup, and stress responses. The evidence does not point cleanly to one single gene as the explanation, and it also does not support treating every deleted gene as an equal 50% loss.

The best overall explanation is that the deletion may reduce biological "reserve" across several systems at once. In this model, many individual effects can be mild, partly compensated, or only become important under stress. That fits the available research better than a single-gene explanation or a simple sum of seventeen harmful effects.

The original question involved photosensitivity and visually triggered seizures. Phase 5 does not prove that connection. Instead, it shows that broader eye and brain phenotypes are currently better supported than photosensitivity itself.

What appears most worth paying attention to

1. Retinal/RPE functional reserve - This is the strongest eye-related signal. Several deleted genes affect photoreceptors, retinal support cells, cilia, phagocytosis, or retinal maintenance.
2. Brain development and neural-network organization - Several genes affect neurites, synapses, neuronal survival, trafficking, or coordinated network activity.
3. Stress-dependent vulnerability - A recurring pattern is that cells may look relatively normal at baseline but become less resilient under injury, oxidative stress, heavy activity, aging, or other challenge.
4. Seizure susceptibility / epileptogenesis - There is some direct epilepsy-related evidence, especially around MIR22, but it does not show that this exact one-copy deletion causes epilepsy.
5. Retinal timing / visual signal processing - Plausible, but not directly measured in the exact CNV.
6. Cortical synchronization / flicker entrainment - Mechanistically relevant, but currently a missing link rather than a demonstrated phenotype.
7. Photosensitivity / photoparoxysmal response - Still plausible, but it sits furthest downstream and currently has the least direct evidence of the major candidate outcomes.

Why one gene is probably not the whole story

Some genes in the deletion have very strong eye or brain effects when they are completely knocked out or heavily damaged. The problem is that the actual CNV is heterozygous: one copy is affected and one copy remains. For several genes, one copy is enough to avoid the severe phenotype seen in full knockouts.

PRPF8 is a good example: reduced RNA does not necessarily mean reduced protein. PITPNA and WDR81 show threshold or recessive behavior, where the severe phenotype appears mainly after much greater loss of function. That makes it risky to claim that any one of these genes automatically explains the whole CNV phenotype.

Why "all the genes add up" is also too simple

Not every deleted gene pushes biology in the same direction. In some experiments, losing TLCD2, MIR22HG, or SERPINF2 can actually be protective under specific stress conditions. RTN4RL1 loss can reduce coordinated neural activity rather than increase it. These findings do not make the genes irrelevant; they show that their effects depend on context.

That is why the leading model is a distributed reserve model rather than a simple arithmetic total of gene losses.

What the simulations actually contributed

The simulations were not used to invent biological evidence. They were used to answer methodological questions: how many biological replicates would be needed, how easily small effects could be missed, how often frequency scans generate false positives, and whether a distributed mechanism could be mistaken for a single-gene mechanism.

- The final standardized simulation suite used 20,000 replicates per scenario/cell where applicable, totaling about 1.88 million primary stochastic draws/resamples.
- With that many runs, Monte Carlo sampling error became very small; the larger uncertainty now comes from biology, not from the number of simulations.
- Small one-copy effects can be difficult to detect with too few independent cell lines or clones.
- Using many cells from the same clone does not replace independent biological replication.

- Scanning many flicker frequencies without correction can produce apparently significant results by chance surprisingly often.
- Small datasets can make a distributed mechanism look like a simpler single-gene explanation.

So what does Phase 5 actually conclude?

The deletion is best viewed as a candidate distributed visual-neural reserve lesion, not as a proven single-gene photosensitivity mutation.

The strongest candidate consequences are subtle retinal/RPE dysfunction, altered neural development or network organization, and reduced resilience under stress. There is also moderate support for an effect on seizure susceptibility or epileptogenesis. Photosensitivity remains possible, but the key physiological chain from the retina through the thalamus and visual cortex to a photoparoxysmal response has not been directly measured for this exact CNV.

What has NOT been proven

- That every deleted gene loses exactly half of its function.
- That the genes act synergistically.
- That the exact CNV causes a measurable retinal abnormality.
- That the CNV causes abnormal visual-cortex excitation or flicker entrainment.
- That the CNV causes a photoparoxysmal response or visually triggered seizure.
- That one specific gene is the sole driver.

Why the result is still useful

The project started with a broad question about photosensitivity. Phase 5 narrowed the problem. It now appears more defensible to study the CNV as a broader visual-neural reserve problem, with retinal maintenance, neural-network organization, and stress response as the

strongest candidate phenotype domains. That is more specific than simply saying "we do not know," while still respecting the fact that the missing physiological measurements have not been made.

The next project stage, Phase 6 - Conclusion, integrates this frozen Phase 5 record with the earlier genomic, mechanistic, and expression phases. Phase 6 is a separate master conclusion and does not alter the frozen Phase 5 findings.